Title of Proposal: MoniTank: Underground Storage Tanks Risk Mitigation System for petrol fuel stations.

MoniTank

Underground Storage Tanks Risk Mitigation System

History of changes

Version	Date	Changes
1.0	27.01.2017	Remark: Passive guided waves, GW (consistency)
2.0	08.05.2018	Remark: Exclusion of guided wave technology, extension of reporting period and project, and use of subcontractor

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1. Excellence

MoniTank Product fast track commercialisation aims at providing continuous condition monitoring system for underground storage tanks applications in order to avoid risk, structural failures and losses associated to tanks leakage and corrosion.

Figure 1 Fuel station explosion due to UST failure

The underground storage of petrol in tanks, especially in close vicinity of urban and rural environment (petrol fuel station), may pose several risks e.g. fire/explosion, environmental damage, health effects when handled by individuals. Because of the large volumes of petrol used daily (only in EU >400 million litres) and large service stations network (>100,000 only in EU) and storage capacity surface and subsurface releases are likely to occur.[1] Petrol gives off highly flammable vapour even at very low temperatures. Because of the **flammability of petrol vapours**, service stations carry a risk of **fire or explosion** not common to other types of retail outlets. Petrol vapour is heavier than air and tends to sink to the lowest possible level of its surroundings and may gather in tanks, cavities, drains, pits or other low points and will travel across the ground due to gravity (down-hill) or may be carried in the direction of the wind. Petrol vapours can find their way into basements of buildings and public drains with serious consequences should the vapours come into contact with an ignition source. If released into the environment, petrol and diesel may also **pollute the soil and water supplies**. Whenever petrol escapes from an underground storage tank (UST) or pipelines, it can travel significant distances.

Leaking underground storage tanks are the most numerous sources of MTBE contamination (Fogg et al., 1998). It is estimated that **10-35% of underground storage tanks in the USA are leaking or fail to pass a tightness test** (Page, 1989). In one survey of **3,500 tanks, only 2% of the tanks were leaking while 10% failed to pass the tightness test**. However, in about **9% of the service stations surveyed petrol was found on the top of groundwater in close proximity to station** (Page report, 1989).Beside that almost 270[2] **accidents** between 1958 to 2007 in France occurred to due to **USTs failure**(ARIA database, Ministère de l'Ecologie, du Développement durable et de l'Energie, FR). The level of risk associated with fuel leakage means early detection of leaks is essential. The very high cost of fuel also means the monetary

[1] http://echa.europa.eu/documents/10162/5602401f-1d75-4931-aa77-abd37d436dc9
[2] www.aria.developpement-durable.gouv.fr, in the "Analysis andFeedback" section.

loss caused by fuel leakage should be of concern to businesses, as a **10,000 litre loss =13,500 Euros[2].** These problem challenges and losses create an opportunity and need of continuous condition monitoring system MoniTank a system developed to address needs of monitoring and inspection of underground storage tanks. SHELL operates 14000 fuel stations[4] and deploy USTs in filling stations in EU has shown interest in a commercial MoniTank system.

Consortium behind this action aims to commercialize **MoniTank system** (already proven at **TRL6**, received positive feedback from potential clients) a low cost monitoring system that actively monitors the underground storage tanks(USTs) online and can flag changes in condition before leakage occurs(**99% Probability of detection (POD)accuracy**). It will offer an innovative system for the early detection of leakages through the assessment of the structural integrity of USTs (tightness, corrosion) in order to predict their maintenance using a combination of Risk Based Assessment (RBA), continuous Structural Health Monitoring (SHM) and Quantitative Risk Assessment (QRA) taking advantage of key enabling technology of acoustic emission. The system will be meet the ASTM and BS standards of inspection and NDT and will be in full compliance to the Environmental Protection Agency (EPA) mandatory UST monitoring protocols. Commercialisation of MoniTank will create **€24 million in revenues**, **€11 million** in **gross profits** and **will create 112 direct jobs** within **5 years of commercialisation for consortium**.

Integrity NDT has developed the system internally with TWI and BUL and demonstrated it at Floteks tanks a commercial manufacturer and supplier of underground storage tanks, the proved system at TRL6 planned to be tailored in this project as per feedback of potential customers mainly filling stations in terms of ease of operation and scaled up monitoring. Consortium has been established by bringing in IRIS an advance engineering SME expert in developing cloud based, IoT enabled system and software to finalise a commercial version of MoniTank at TRL9, with the aim to enter market within 3 years of start of this project.

1.1 Objectives

- **Specific Objectives:**

Specific Objectives of MoniTank Project are as follows:

Table 1: Specific Objectives

Scale up, Testing and Demonstration Objectives		
Specific Objectives	**Tasks**	**Deliverable / Means of Verification/WP**
MoniTank system scale up to commercial version considering commercial value, manufacturability and end user requirements. *(by Month 16)***TRL 7**	System optimisation, scaling up performance, communication system and signal processing	Scaled up hardware for commercial version *(Activities to be done in WP1)*
Risk based inspection algorithm fine tuning and defect recognition software with updated graphical user interface(GUI) *(by month 16)* **TRL 7**	Full integration, bug fixing and user friendly GUI update	Fully functional integrated GUI interface Software of MoniTank system *(Activities to be done in WP1 & WP2)*
Validation of MoniTank predictive accuracy(Pre-Production commercial ready product) *(by month 28)***TRL8**	Build reporting functionality within the MoniTank software, allowing for bespoke as well as standard reports to be produced	Fully functioning commercial system *(Activities to be done in WP2)*

[2] Class 1 LeakDetection - The ultimate protection, PetrolPlaza Technology Corner (August 2012, Thompson Jamie) [4] http://www.retail-index.com/Sectors/PetrolRetailersinEurope.aspx

Minimisation of MoniTank's system costs and market ready MoniTank Product packaged system *(by month 30)***TRL 9**	Industrial testing, Site and field trials and demonstrations at commercial scale	All trials and tests completed in Floteks USTs for three month and results analysed *(Activities to be done in WP1 & 2)*
Certification of the MoniTank system*(by month 28)***TRL 9**	The system should obtain the CE marking, ISO/IEC certification and the API 581 (Plants operating Standard)	Certification: allows MoniTank users to comply with legislation and inspection protocols.
Commercial Objectives		
Specific Objectives	**Tasks**	**Deliverable / Means of**
Production of Training Manuals, User Guides and Training Programmes (*by month 24)*	Development of full training collateral to support the MoniTank product usage.	Training Materials produced, training courses developed *(Activities to be done in WP3)*
Launch of online advertising campaign, marketing and sales strategy implementation and pre-launch clients engagement *(by month 30)*	Developing online marketing campaign to build audience before launch, development of complete sales and marketing plan before and after launch of MoniTank	Advertising material developed, campaign prepared and customers engaged before launch*(Activities to be done in WP4 & 5)*
Development of full commercialisation roadmap and sales agents network extension*(by month 30)*	Produce full commercialisation and investor ready business plans. Sales agent network extension and product training	Draft and final business plans produced. Commercialisation plan including pre-contract agreements *(Activities to be done in WP4 & 5) secured)*
Secure initial contract for sales of MoniTank *(by month 30)*	Engage with Tier 1 customers during the final phases of the action to secure contracts for MoniTank to be implemented following the completion of the	Engagement with Tier 1 customers undertaken Pre-agreements in place 2 sales Contracts secured *(Activities to be done in WP 5)*

Project Outcome: Continuous sensor based condition monitoring system capable of transmitting condition of underground storage tanks online and on defect recognition software in hands of operators making it possible to inspect underground storage tanks without need of decommissioning or downtime monitoring activity.

Novelty of Product: Automatic and continuous condition monitoring system based on acoustic emission for structural health assessment of underground storage tanks in petrol filling stations and oil/gas production facilities integrated with digital technology of Internet of things gateways, cloud based data collection and storage, wireless sensor network and innovative risk based inspection algorithm and software for informed decision and preventive maintenance of USTs.

Problems and Project relations: In Underground storage tanks most damaging leaks are likely to occur with tank failure. The most common cause is corrosion. Most tanks are steel, which has a natural tendency to corrode. Corrosion is a natural electrochemical process, and any factor in the design or installation of a tank system that enhances this process can result in accelerated corrosion and premature tank failure. For example, corrosion may be accelerated by the following: the use of corrosive or nonhomogeneous backfill; the use of dissimilar metals in the tank or piping; the presence of shallow groundwater or temporary water around the tank; damage to corrosion control coatings; and stray currents from electrical power sources. Leaking USTs are one of the leading causes of groundwater contamination.

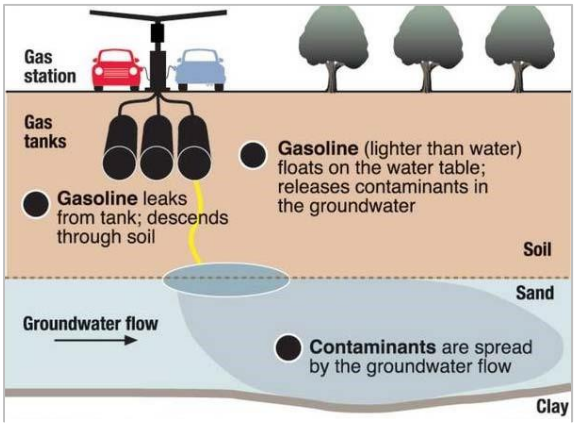


Figure 2 UST failure contamination underground

They pose the greatest threat in rural areas, where normally 95 percent of the population depends on well water for drinking in EU. As a result, local governments, businesses, and private citizens face substantial costs to comply with regulations covering both the tanks and potential clean-up, in case there is a problem. Where the average cost for cleaning up a spill or leak is more than €70k. **Detailed surveys, interviews done by Integrity NDT and TWI with SHELL, TOTAL, BP, ESSO, ENI that operate majority of stations across EU and worldwide has expressed an interest in MoniTank as currently, there is a lack of robust continuous condition monitoring system**.

Table 2: Challenges vs. Proposed Solution

Challenges to be addressed	Project solution and value innovation
Industrial: Increasing transportation and logistics needs result in having 132k filling station only in EU increasing at 3.5% CAGR containing more than one underground storage tank at a station will require continuous monitoring system for critical fuel storage tanks for performance and cost efficiency.	MoniTank will increase reliability and performance of USTs at fuel stations making them more secure, and cost efficient in terms of operation and as a result of reduced cost of inspection overall by 75%. MoniTank will be able to detect all structural faults in tanks and will transmit monitoring data via sensor network to central cloud based software that will show risk profile, faults and maintenance priority areas of tanks.
Economical: The level of risk associated with fuel leakage means early detection of leaks is essential. The very high cost of fuel also means the monetary loss caused by fuel leakage should be of concern to businesses, as a 10,000 litre loss =13,500 Euros[3].	MoniTank will reduce cost of inspection by 75% and will minimise the chances of USTs failures un detected by more than 90% providing a preventive maintenance avoiding leakages and fuel losses.
Social: Risk of incidents due to leakage or explosion in underground storage tanks can affect citizens on fuel stations.	MoniTank system will enhance safety as at any time operators of fuel storage tanks can see the condition of tanks and avoid any leakages or failures before occurrence.
Legislative: EPA and environmental regulatory authorities make it mandatory to have continuous monitoring of duel storage tanks underground.	MoniTank will be in compliance with standards and legislations by providing 99% accurate probability of defect recognition more than as asked by compliance of monitoring system for USTs(98% is requirement).

Relevance to Horizon 2020 and EU wide Impact: With three industrial companies in the consortium (Integrity NDT, IRIS, FLOTEKS), the MoniTank action will directly address the European need to support SME's to innovate in the market place. Integrity NDT will use the MoniTank product to rapidly accelerate growth in the company and the introduction of further new products, based on European IP and manufacturing in Europe. For IRIS, the action will enable access to new markets in the oil and gas industry based on proven integration technology. The main beneficiaries will be European companies. Oil and gas production and storage plant operators and provision of maintenance technologies are major European industries exporting technology into global markets with much of the infrastructure and engineering based within Europe. Through this direct impact on the European economy, the action addresses H2020 priorities of encouraging SME participation and economic growth as well as priorities in securing safer, cleaner, more secure energy product supplies. MoniTank addresses the EU Societal Energy Challenge, which aims at supporting "the transition to a reliable, sustainable and competitive energy system"[4]. The effective maintenance of ageing infrastructure using the MoniTank will reduce leakages and damages in storage tanks in more efficient energy provision and a safer and more secure environment. The wider European Impacts and relation to H2020 objectives are summarised in Table 3.

Table 3: European Impacts and Relation to H2020 Objectives

[3] Class 1 LeakDetection - The ultimate protection, PetrolPlaza Technology Corner (August 2012, Thompson Jamie)
[4] http://ec.europa.eu/programmes/horizon2020/en/h2020-section/secure-clean-and-efficient-energy

Employment	**MoniTank** will create new employment opportunities (**112 new jobs within 5 years**) for the European consortium and those involved in inspection of UST structures in fuel stations with high potential for future growth.
Fuel stations and public safety	**MoniTank** will strengthen energy safety by ensuring the integrity of tank structures, and guaranteeing supply of fuel/petrol/gas without disruptions or threats to environment, health or life.
Meeting increasing energy demand	MoniTank will allow more cost effective operation of USTs with reduced inspection cost, increasing available capacity and throughput.
Innovation	Introduction of **MoniTank** will cement the reputation of the commercial partners in the consortium for bringing to market innovative new products for this important industrial market, allowing them to finance and sell further new products and services. This will reinforce the position of European industry, and the strong innovation capacity of European SMEs, to bring new technologies to market and remain competitive in the global market, ahead of strong competition from the US, Asia Pacific and Middle East.

Table 4: Project solution and associated values

How Project outcomes solve stated Problem	Buyers Value(End Users)	Business Opportunity Sellers Value (Consortium)
MoniTank will provide an automatic structural monitoring platform for underground storage tanks failure monitoring which represent the most number of failures in UST structures. By preventive maintenance and monitoring data of such tanks maintenance operations will be scheduled before failure or any damage occurs ultimately, enhancing fuel stations operations productivity and reliability, reducing failure risks and protecting environment, public and workers.	Tank operators will be able to ▪ Increase safety by75% ▪ Reduce downtimes and maintenance cost by 60% ▪ Reduce Failure Risks by 90% ▪ Significant improvement in USTs maintenance compliance ▪ 90% reduction in inspection time	MoniTank commercialisation will create for consortium €24 million revenues; generating €11 million gross profits and creating 112 jobs over 5 years post action completion.

1.2 Relation to the work programme

Relation to topic of the developed commercial product is as follows:

Table 5: Topic expectation vs. Project relevance

Topic: Fast Track to Innovation of the Work Programme	
Topic Expectation	**How Project addresses the stated expectations**
Technological innovation starting at least at TRL 6	MoniTank prototype has already been developed, tested, validated and demonstrated in a relevant environment on actual underground storage tanks. This work has been undertaken through investment allocated by private and public sources The system's performance proved the high business potential and prompted us to follow through with additional investment in order to fully commercialise it.
Relation to "Leadership in enabling and industrial technologies" and/or to any of the specific objectives under the priority "Societal challenges"	MoniTank, as explained above, is set to greatly reduce accidents and contribute to increasing the safety and operation of storage tanks in fuel station and process industry while advancing cost-effectiveness by reducing the maintenance costs. Thus it is perfectly in line with the highlevel EC objectives of Industrial leadership offering a competitive, cost effective structural health monitoring system with a strong potential of generating export revenues and competitiveness of EU companies involved in its commercialisation.

Clear and detailed business and market analysis	We have formulated a detailed business plan founded upon market research. As presented in Section 2.1 our plan addresses the market structure and competitive landscape while articulating the IP exploitation process and the resulting economic benefits for each member of the value chain (as shown in section 2.1) and the course that will be followed in delivering the product to market from manufacturing to distribution.
High Industrial participation	Our consortium is industry dominated, as it includes 2 **industrial partners** forming a value chain set to deliver a quick and successful market take-up for MoniTank. **INT NDT,** are committed to seeing the commercialisation process through leveraging their established positions and their supplier connections across Europe. Moreover, **TWI and BUL** have a strong network of industrial alliances and members to leverage the sales of MoniTank. **Floteks** is one of the large end user, they will be the potential client for MoniTank system and it's testing during validation process. **Floteks will endorse the MoniTank System** and will be the first customer to buy the commercial system.

1.3 Concept and approach

Overall Concept and Methodology:

The concept of MoniTank is shown in figure. Acoustic Emission (AE) techniques is proven for screening pipelines for corrosion and erosion. This capability will be extended into condition monitoring of the UST by permanent installation of the transducers.

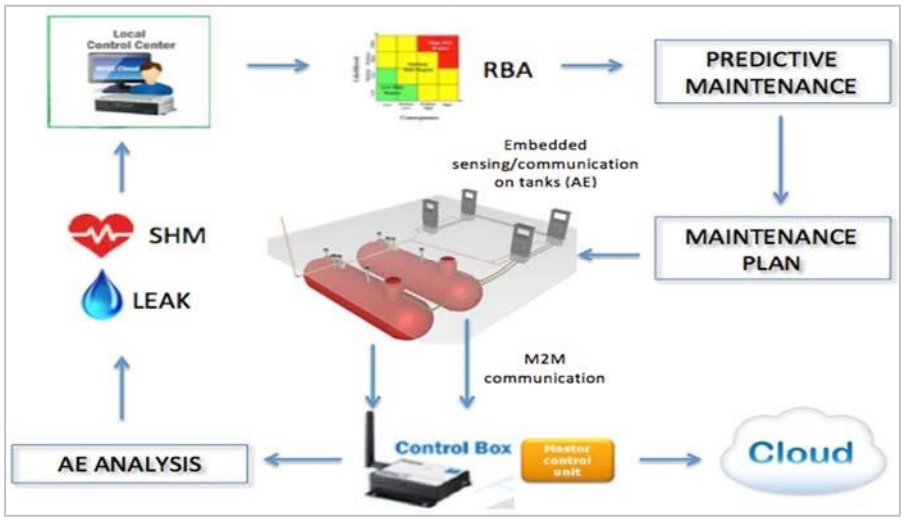


Figure 3 MoniTank Product Concept

This Project aims to commercialize, MoniTank a low cost monitoring system that actively monitors the tank online and can flag changes in condition before leakage oc innovative system for the early detection of leakages through the assessmer structural integrity of USTs (tightness, corrosion) in order to predict their mai a combination of Risk Based Assessment (RBA), Structural Health Monitorin Quantitative Risk Assessment(QRA). The data coming from the AE sensors wireless to the master control unit and collected and analysed in the control continuous risk-level classification (EN15856) allows immediate notification or critical levels are reached. This automatic fault recognition allows operato failures and leakages before they happen and to predict the maintenance of tanks optimizing the maintenance plan. A method for online condition-based monitoring (CBM) providing a diagnostic and predictive tank model based on historical data an operation data.

Figure 4 MoniTank Product trials

This SHM process will be low cost using a small number of sensors, and wil indication of the condition of the tank without the need for any invasive inspe The proposed SHM strategy will combine liquid level and corrosion monitori detection of sedimentation. A simplified solution to SHM of buried storage ta an acoustic emission based strategy will be implemented. Table below shows typical inspection of tanks and associated cost, where MoniTank can reduce 75% of the cost by providing a standalone system of monitoring.

Figure 5 MoniTank Demo site

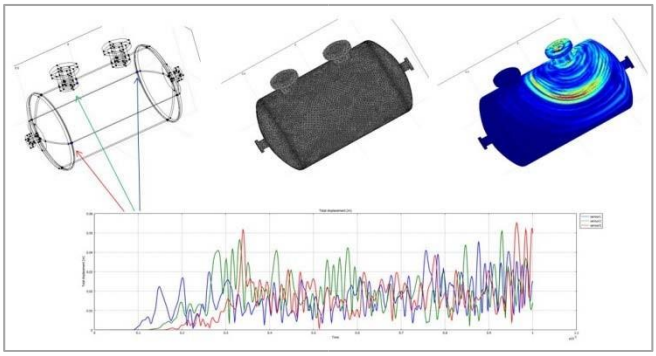
Table 6: Typical USTs inspection process and associated cost

Process Step	Description	Estimated cost (€)
1	Prepare recipient Tank (tank that will store the product until inspection is finished)	30,000
2	Move products to recipient Tank	30,000
3	Open and degas tank	60,000
4	Clean tank (labour- and capital-intensive process requiring equipment)	120,000
5	Non-Destructive Testing - Magnetic Flux Leakage technique (semi-automated)	70,000
6	Non-Destructive Testing - UT technique (manual)	60,000
7	Close and Seal tank after NDT is carried out	40,000
8	Refill tank with product (from recipient tank)	30,000
9	Check seals, vents, hoses etc.	10,000
10	Paper work, safety checks, management & supervision of above process	10,000
The above process will typically last for about four weeks and incur an overall cost ranging within €500k to €1m		

Key Enabling Technology application to create commercial value of MoniTank System:

Key enabling technologies considered to create commercial value of MoniTank system are as follows: **Table 7: KETs vs. Commercial Value**

Figure 6 MoniTank System validated at TRL 6



Commercial Value/ Business Targets	Technology Enablers	
	Automated Continuous Condition Monitoring	**Optimised Preventive Maintenance**
Safety	Automated continuous condition monitoring will improve safety of fuel stations and environment.	In service failures will be eliminated.
Performance & Reliability	Continuous monitoring of USTs will improve performance and reliability of filling stations.	Targeted maintenance will reduce variability in service.
Cost Efficiency	Condition monitoring of underground assets will improve availability and reduce cost ultimately enhancing efficiency.	Failure prediction will reduce asset cost of ownership.

A **prototype (TRL6)** has already been developed and tested with the transducer permanently mounted on a UST.

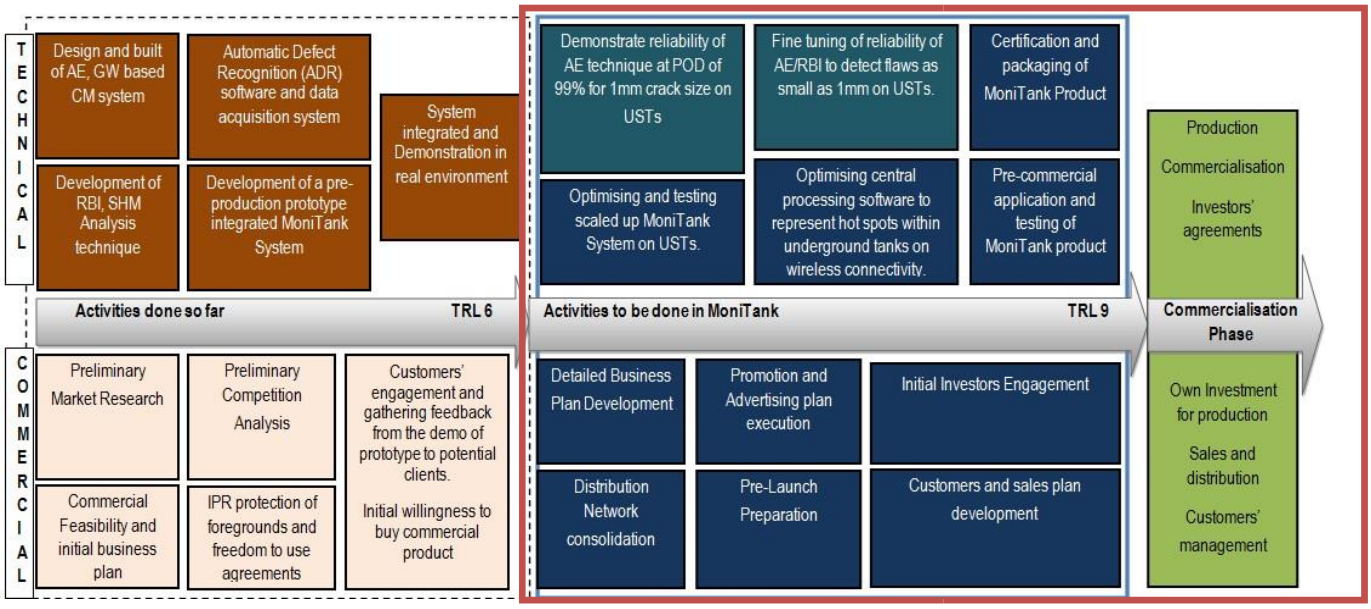
The following features and enabling technologies has already been developed as shown in figure above:

- Validation of the choice of the sensors including the validation of either: the conventional piezo-electric or the flexible fibre composite.
- Signal processing routines to improve the signal to noise ratio.
- The transducers have been installed permanently on a UST. This raised specific requirement with regard to the long-term ultrasound coupling to the tank wall, and the packaging of the transducers.
- Assemble of the electronics. TWI's BIP has been utilised to deliver the hardware prototype which required specific software development to drive the transducer arrays and to meet the site conditions.

Once the UST structure, specifications and techniques have been analysed, a specific array of sensors has been designed as well as the selection of the most appropriate hardware:

- The prototype comprises multichannel AE capabilities for characterization and localization of acoustic emission sources with signal filtering within the frequency range 50-250kHz;
- The prototype functions in normal operation at frequencies between 10kHz and 250kHz.

Project Positioning: Activities done so far and to be achieved in this project are listed below:



Measures in concept development for effective market take-up:

MoniTank product development measures planned in this project for effective market take-up include following tasks:

- **Condition Monitoring & Inspection Cost Reduction:** System performance to optimise to achieve inspection cost reduction by 75%, inspection process time by 60% and inspection labour time by 40%. (*Activities in WP1,WP2*)
- **Cost Reduction of the System:** In order to optimise the system cost during scaling up the system, a detailed components/suppliers/parts availability analysis will be considered, and most cost effective, easily available and high performance components will be used to finalise commercial version of MoniTank. Supply chain will be sorted for manufacturing of MoniTank carefully considering cost and value. *(Activities in WP 4)*
- **Enhancement of commercial value of the product by extending the performance and application of the system:** 100% automatic , continuous condition monitoring, 99% defects recognition accuracy and long lasting sensor network functionality*(Activities in WP1,2,3)*
- **Compact and portable design:** Intrinsic safe electronic packaging will be developed in MoniTank weighing overall system offering a compact system for operators for ease of use and operations. *(Activities in WP2)*
- **Easy to operate system with user friendly GUI:** To enhance market uptake of MoniTank system for customers, a user friendly GUI, with detailed product manual, training guidelines and user/operator training on sale of system is planned in project concept development as well. *(Activities in WP1,3)*

European dimension of proposed Innovation and added value for Europe:

MoniTank system will address EU-wide challenges by creating more job opportunities in NDT sector. It will offer an innovative solution for quality inspection of underground tanks in all kinds of plants specifically in fuel stations before occurrence of any crucial problems avoiding accidents and losses thus; will contribute to the environmental and workers safety. The system will be distributed and sold to global markets as detailed in section 2 resulting in increasing export revenues for Europe as well contributing to EU competitiveness and economic impact.

Table 8: EU Impact and Relevance to H2020

Job Creation	MoniTank will create new employment opportunities (**112 new jobs** within **5 years between consortium members**) for the European consortium and those involved in NDT inspection of tanks. The solution provides opportunities in other industries (petrochemical, chemical etc.).
Environmental Safety	MoniTank will strengthen energy safety by ensuring the integrity of tanks, and guaranteeing continued fuel stations operations without neither disruptions nor threats to health or life. MoniTank will be in compliance with **ASTM, BS standard**

	about ensuring the integrity of process tanks systems in the Oil & Gas industry is a key issue with respect to the process, business, safety, and the environment[5].
Energy and cost efficiency	MoniTank will directly impact energy cost by protecting fuel storage tanks that are vulnerable to leakage, loss of fuel, corrosion and safety of filling stations.
Innovation	Introduction of MoniTank technology will cement the reputation of the commercial partners in the consortium for bringing to market innovative new products for this important industrial market, allowing them to finance and sell further new products and services. This will reinforce the position of European industry, and the strong innovation capacity of European SMEs, to bring new technologies to market and remain competitive in the global market, ahead of strong competition from the US, Japan and China.

Gender Analysis:
MoniTank will develop and implement a Gender Action Plan to promote gender equality in all possible forms within the project ensuring gender equality, promotion of women recruitment and maintaining gender balance.

1.4 Ambition

Through commercialisation of MoniTank product we aim to create **€24 million in revenues**, **€11 million** in **gross profits** and **will create 112 direct jobs** within **5 years of commercialisation for consortium**.
State of the art: Currently, the most advanced approach to tank inspection is by internal inspection using manual non-destructive testing (NDT) techniques. This requires removing the tank from service, removing the contents, then venting and cleaning it to make it safe for manual access. Existing monitoring systems on the market such as manual gauging, inventory control and leak detection systems only rely on the tank leaking before it can be detected. The EPA requires the tank to be monitored for a leak at least once every 30 days using interstitial monitoring[6]. For many years in Europe the most utilized *Secondary Containment with Interstitial Monitor* has been the class 2 liquid system (operating using a liquid filled interstitial space) but there has been a move in recent years by environmentalists and responsible at oil companies to the Class 1 system (using air or nitrogen pressure or by pulling a vacuum on the interstitial space).
This change is based on a variety of reasons:

- In environmentally sensitive areas, even the leak detection fluid is considered a hazard to the groundwater
- When leaks occur the detection fluid in the interstice has to be disposed of
- Economy - the cost of leak detection fluid adds to the cost of tank manufacture
- Operators are taking environmental issues seriously and are changing to the more environmentally friendly systems

Pressure and vacuum leak detection systems for double wall tanks and pipes have been operating in Europe for over 35 years. There are over 700,000 such installations in use monitoring tank and pipes. They detect a small number of leaks each a year, around 1% of those being monitored. 75% of those leaking have a leak in the outer wall and 25% in the inner wall. There are no existing standalone monitoring system capable to detect leakage in both outer and inner wall of underground storage tank as proposed innovation in MoniTank.

Table 9: Beyond state of the Art and MoniTank Product advantages

MoniTank Technology	SOTA (State Of The Art)	Beyond SOTA; Product Differentiation	Product Advantages Related to Market
AE sensor platform and RBI techniques	Sensor separate from the processing functionalities.	MCU integrated into the smart sensor platform	The AE sensors can send a result and can be configured for different CM application.
Wireless communication	Very well established	Wireless integration to direct transmission to the cloud	Send results or data to the cloud

[5] file:///C:/Users/InnoTecUK/Downloads/process_piping_guide_R2.pdf
[6] https://www.gpo.gov/fdsys/pkg/FR-2015-07-15/pdf/2015-15914.pdf

Cloud for Condition Monitoring system	Not used yet	CM data travelling by the cloud	Data available to correlate results from different sensors from different sites.
CM software to cloud management	Not existing for CM systems	Development of a new type of software	Data management from different sites and costumers.

There are more visual inspection NDT services available on market for USTs inspection and there is lack of systems for condition monitoring of underground tanks which makes MoniTank an attractive product to capture not very much saturated market. **Table 10: Competitive Overview**

Product and Service on Market	Limitations	Differentiation and value added benefits of MoniTank
SGS ,UK ▪ Conventional internal inspection: Magnetic Flux Leakage / Ultrasonic Testing	Typical NDT service requires decommissioning, downtime and high labour cost of inspecting tanks,	▪ MoniTank is a standalone condition monitoring system capable of detecting structural flaws in underground storage tanks with 99% POD, capable to monitor metallic double wall structures of tanks, will use cloud based communication and data acquisition system so will be easy for operators to monitor and inspect condition of underground tanks anytime without extra inspection time or labour cost. ▪ MoniTank will reduce cost of inspection significantly by 75% as once installed on tanks sensors will transmit condition of tanks which will be analysed using advance risk based inspection techniques to give a detailed risk curve for operators and maintenance personal.
MISTRAS,USA ▪ TankPac Acoustic Emission Inspection for Tanks ▪ Manufacturer of Acoustic Emission testing equipment	Limited output of inspection only detects condition of structure walls and not able to provide high thickness low size defects data.	
BMS Group, UK Veeder-Root is the world's leading supplier of automatic tank gauging and fuel management systems and one of the few companies in the industry to design, manufacture and service its own products.	Not offering a complete standalone system but offer different monitoring gauges, not able to detect corrosion and internal tank faults.	

Table 11: Techniques comparison

Analysis/ Technique	Application	Limitations	MoniTank Technique relevance for USTs condition monitoring
Infrared Thermography	Infrared thermography (IRT) has become a matured and widely accepted condition monitoring tool where the temperature is measured in real time in a non-contact manner.	Limited to temperature based monitoring, more relevant for heated objects condition monitoring.	MoniTank is based on acoustic emission technique. AE technique emanates from the practice of sorting good parts from 'clunkers' by hitting them and listening to them ring (hence other commonly used related defects that have led to metal to metal contacts (i.e. lubrication breakdown). AE uses higher frequency detection. Metallic USTs in its working condition produce acoustic signal and this signal is a characteristic signatures of that structure which does not change over time. However, in cases of structural or functional anomaly or failure, the dynamic characteristics of the USTs change which is reflected in its acoustic signals. The
Ferrography	Ferrography is a technique for analysing the particles present in fluids that indicate mechanical wear.	Limited to fluid based condition monitoring.	
Ultrasonic condition monitoring	Ultrasonic condition monitoring (UCM) is a technique that uses airborne (non-contact) and structure borne (contact) ultrasound instruments to	High frequency, complex and costly condition technique used for large application and systems.	

Analysis/ Technique	Application	Limitations	MoniTank Technique relevance for USTs condition monitoring
	receive high frequency ultrasonic emissions produced by operating equipment, electrical emissions and leaks etc. to monitor the condition of equipment under test.		nature of the developing fault has unique acoustic characteristics which can be compared with the acoustics signatures of the tanks working under normal operating condition. By using innovative signal analysis algorithm developed in MoniTank device, it is capable to determine the exact category/type of fault and then transmit the data wireless to the cloud platform that can be accessed on site, central control unit or any other location remotely to analyse underground tanks performance.
Lubrication oil Analysis	Lubrication oil is used in electrical and mechanical machines to reduce friction between moving surfaces. The lubrication oil is important source of information for early machine failure detection.	Lubrication oil based condition monitoring, limited to particular application where lubrication oil is used.	

Table 12: Expected key market applications and differentiation factor for Potential customers

Project Result/outcome	Key Market Application	Differentiation Factor(Added value for potential customers)
Structural health condition monitoring system for underground storage tanks	-Fuel/Petrol Station -Chemical Plants -Process Industries -Storage warehouses Etc.	▪ Cost Effective and 99**% POD** accurate system ▪ Remote access to condition monitoring data ▪ Detailed risk profile and failure localisation ▪ No need to shut down tanks operation ▪ Wireless sensor network and flexibility to place point of interest in tanks

Patent search shows there are no standalone systems available, some companies have developed set of different system to monitor condition of underground tanks but MoniTank will offer one system capable to detect all structural faults in USTs.

Product Lifecycle and Positioning:

Initial developments in project as shown in the figure depicts higher cost of innovation and low sales, but with time and complete market transfer of MoniTank the sales and profit will rise as detailed in section 2 and cost will be reduced due to production of the system. Some of the profit will be re-invested in further, development of MoniTank2 second generation product for competitive and sustainable advantage. Integrity NDT with partners will continuously seek innovation aspects in the product obtaining feedback of clients and their needs and tailoring the system for sustainable competitive advantage.

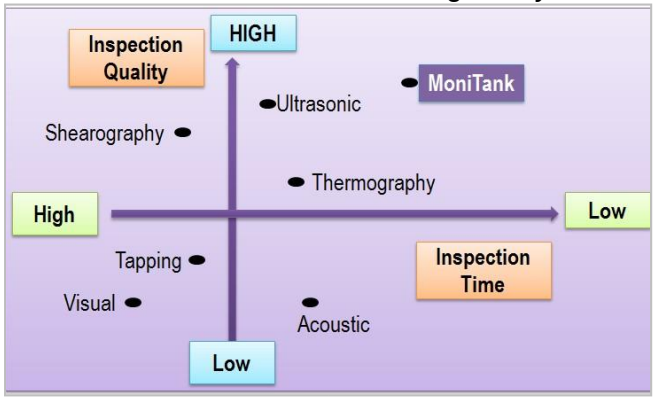


Figure 8 MoniTank Product Positioning

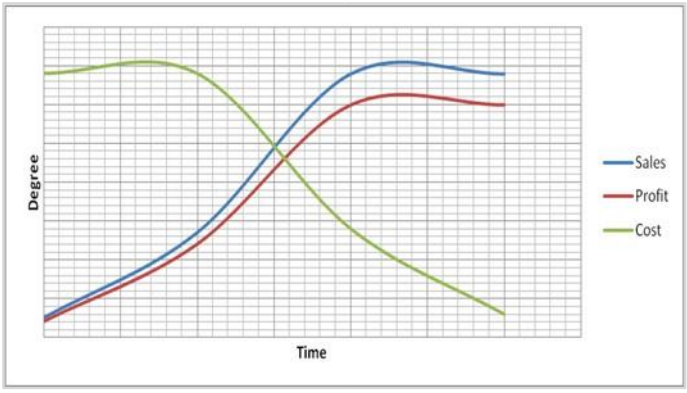


Figure MoniTank Product Lifecycle

2. Impact

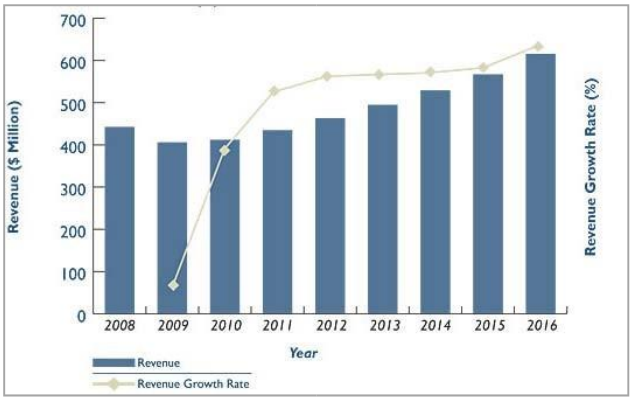
2.1 Expected impacts

Expected Impact under the relevant Topic:
The expected impact of MoniTank Project in regards to topic is as follows:

Table 13: Impact expectation and project relevance

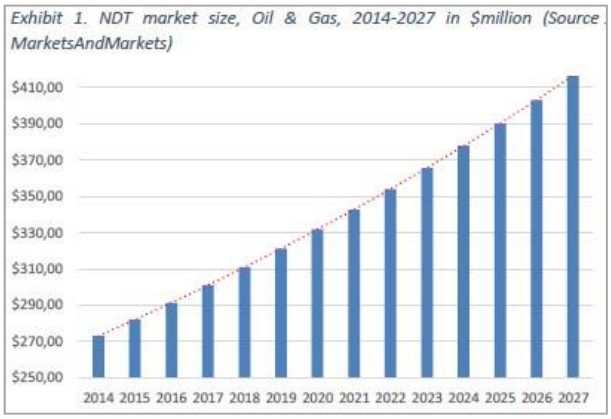
Expected Impact under the relevant topic	Project Impacts
"Fast development, commercial take-up and/or wide deployment of sustainable innovative solutions in enabling and industrial technologies and/or for tackling societal challenges."	The project foreseen a fast development, commercial take-up and wide deployment of an innovative solution to increase the safety of petrol stations and reduce their impact on the environment by detecting in real time the risk of leakages and fire/explosion. It will solve social challenges by reduction of risks and accidents related to leakages and vapour emissions, by providing guideline to oil companies and local authorities in determining the safe boundary of petrol station and improving confidence in petrol stations security and in particular in unmanned/virtual assisted stations.
"Time to initial market take-up no later than 3 years after the beginning of the FTI project."	Based on the market and competition research, we have developed an extremely detailed business plan targeting one of the biggest sectors in the world's energy market and we are perfectly positioned to penetrate it. The project will last for 30 months and the time to initial market takeup will be within 3rd year after the beginning of the FTI project.
"Enhanced competitiveness and growth of business partners in the consortium, measured in terms of turnover and job creation."	MoniTank will enhance the competitiveness and growth of partners by generating €24 million revenues, €11 million in gross profit and creating 112 direct jobs within consortium.
"Leveraging more private investment into research and/or innovation. And Increased industry participation, including SMEs, and more industry first-time applicants to Horizon 2020."	Our project will bring together diverse competences and backgrounds from the research (TWI, BUL) to the industrial (Integrity NDT, IRIS) to the commercial (Floteks) field. Our consortium includes 2 industrial partners with solid positions in the condition monitoring market, 2 RTOs and 1 Underground storage tank manufacturer Floteks.
"Where appropriate, addressing transnational value-chains and/or EU-wide or global markets."	The project addresses transnational value-chains and targets one of the biggest sectors in the world's energy market.

Project Innovation meeting needs of European and Global Market needs: MoniTank product commercialisation meet the technological and economical needs of EU and Global markets, as shown in figure 9 the market for NDT systems have risen to 600 million dollars in 2016 globally and it is still growing which creates MoniTank system relevant in NDT equipment market. Further, NDT market is being segmented on the basis of NDT application (e.g., flaw detection), technology (e.g., AE technology), end-users (e.g., Oil & Gas), and geography. The surface techniques are an established and matured market, with ultrasonic testing having an increasing usage as it provides the advantage of indicating flaw depth. The Oil & Gas industry specifically is highly dependent on NDT to ensure quality and maintain a safe operating environment and remains the largest end user for the NDT market. The global NDT market for the Oil & Gas industry was $273 million in 2014 and is estimated to grow at a compounded annual growth rate of 3.30% to reach $331.7 million by 2020. Figure 10 shows the estimated market size of NDT in the Oil & Gas market from 2014 to 2027 as shown in figure. Growth of the global NDT market, both in terms of production and maintenance, is driven by (1) Stringent government safety regulations. In Oil & Gas industry, an increasing number of assets need to be inspected in order to



prevent any hazards due to leakage and spillage of oil or failures. (2) Ageing infrastructure in developed countries and resultant
deterioration issues. There is a need to extend the usual life of existing infrastructure in order to prevent it from deteriorating further. The average life of assets such as tanks in Europe is estimated to be 40 years; still these assets are required to run at high capacities. (3) Growing demands for NDT from the endusers to optimise maintenance and ensure safe and efficient operation. Regular maintenance of tanks, through their usable life, is required in the Oil & Gas industry to avoid unexpected and catastrophic failures due to fatigue.

Figure 9 NDT equipment Market Trend and Growth



However, a number of facts that impose significant challenges on the NDT market growth, the most important are (1) Impact of macroeconomic shocks. A major hurdle by most NDT companies is providing high quality service at competitive prices. The end users of NDT are highly price conscious; hence, it becomes difficult for NDT companies to provide quality service at lower prices. This is exaggerated in the event of macroeconomic shocks that affect the economy of the country and thus their spending capacity, such as in Asia-Pacific locations where end users are extremely price sensitive. (2) Lack of skilled labour. The lack of availability of skilled technicians is the biggest challenge affecting the NDT market. Considering this fact, it is clear that the number of people taking up NDT inspection has significantly lowered over time. Stricter government regulations and increasing sophistication
of NDT methods has led to an increase in demand of skilled inspectors.

Figure 10 NDT Market Trend and Growth

This demand is currently outweighing the supply. Since the NDT industry relies on skilled technicians for revenues, their shortage poses significant problems for growth. Global growth of storage tanks by regions is shown in figure 11. MoniTank Project will create a positive and economical impact in meeting needs of EU and global NDT market needs as well as storage tanks inspection problems and challenges.SHELL and BP has shown intentions of buying commercial MoniTank System as well as Malaysian university wants to deploy the system for fuel stations in Malaysia run by government.

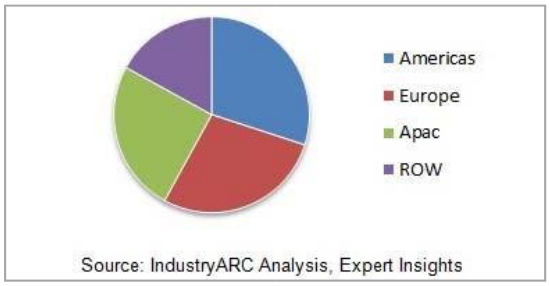


Figure 11 Global Underground Storage Tanks by Regions

Competitiveness and growth of Industrial legal entities in the consortium:

The consortium members will benefit in terms of growth due to a higher level of competitiveness from the innovative uniqueness provided by the
MoniTank system. The integration of new knowledge from the action will be as a result of (i) enhanced monitoring system to monitor the condition of underground storage tanks. Scaling up the system will lead to (ii) Evolved condition monitoring platform. The System platform will be developed to incorporate a full reporting mechanism which will have the added functionality of being user bespoken, depending upon the requirements of individual operators. Through the delivery of this action and the commercialisation of the system, the consortium members will experience knowledge growth as well as commercial/business growth.

Improving innovation capacity and integration of new knowledge:

MoniTank will overall enhance and improve the innovation capacity of partners and will commercialise the system where there is a limited research work available. Partners including **IRIS, INT NDT and TWI** will be able to enhance their innovation capacity in engineering solution development, **BUL** will be able to enhance its services and knowledge, **FLOTEKS** will endorse the system and will enhance its operational performance of storage operations.

Environmental and Social Impact:

Environmentally MoniTank offers a tool to ensure safety and security of underground pressure vessels and through preventive maintenance will secure the environment from damages that can occur due to tanks rupture or breakdown ultimately not only protecting the environment but also the soil from hazardous exposure.

Table 14: User Needs vs. Project outcomes:

User Needs	MoniTank solution	Differentiation factor with state of the art methods
▪ Reduce risk of catastrophic failure. ▪ Reduce risk of low-level leakage and environmental damage ▪ Reduce environmental impact from leaks ▪ Enable compliance with EU directives (SEVESO III Directive 2012/18/EU, Directive 2006/11/EC, Dangerous Substances Directive) and local safety and environmental regulations.	Sensor fusion enables continuous inservice monitoring for rapid early warning of problems and forecast data to ensure timely maintenance	Visual or ultrasonic inspection is of limited effectiveness, particularly floor bottom corrosion. Internal inspection can only be done infrequently because of cost.
Demonstrate compliance with EU directives (SEVESO III Directive 2012/18/EU, Directive 2006/11/EC,	Cloud-based service automates collection, analysis, archiving, and presentation of data for efficient	Current manual inspection records are error-prone and time consuming to manage especially over a 20 year lifespan of a tank
User Needs	**MoniTank solution**	**Differentiation factor with state of the art methods**
Dangerous Substances Directive) and local safety and environmental regulations.	reporting and auditing of safety and environmental management.	
Co-ordinate and share information with other sites, other operators, and government agencies to share best practice and experience and meet new SEVESO III requirements	Cloud-based platform for data management allows tailored access and sharing of inspection, maintenance and incident information to improve risk assessment and spread best practice.	One of the main causes of accidents is poor communication and coordination within the company and between organisations. Currently inspection and maintenance data are held locally with no off site access.
Improve business efficiency through accurate scheduling and coordination of internal inspections and maintenance	MoniTank sensor fusion provides improved forecasting accuracy for timely scheduling of intervention to a maintenance plan that minimises risk and makes efficient use of test and maintenance resources	Outside of infrequent time-based inspection, maintenance is done reactively as problems occur. This increases risk of environmental damage or accidents and makes scheduling of maintenance difficult.
Improve business efficiency by integrating inspection and maintenance into site management processes	MoniTank provides systemised records of inspection results and forecasts, available enterprise-wide.	Current manual inspection records are kept locally and not integrated with site management systems.
Reduce costs of inspection and maintenance programmes	Performing costly internal inspection and repair only when needed can provide cost savings of €50 000 p.a. per tank.	Tank inspection based on time alone means that many internal inspections (€500 000 per inspection) are unnecessary or premature.

Targeted Market (size, trends, and growth rate) and Project Consortium Strategy: There were over **132,136 petrol stations in the EU**, Norway, Switzerland and Turkey operating in 2014, fuelling some 230 million cars and over 30 million trucks on Europe's roads. The USA had 114,474 filling stations (gas stations) in 2012, according to the U.S. Census Bureau. NDT Inspection in Oil and Gas, currently rated $8,1 billion, is expected to grow from 2015 with a global CAGR of 4,6% to reach $10,2 billion[7] in 2020. We believe economic growth will push this number even higher. Most of the oil and gas structures are owned and operated by large oil and gas companies and a great majority of these companies outsource their inspection requirements to SME inspection companies. The inspection market in Europe is dominated by SMEs.

By overcoming the safety issues and data limitations of current inspection techniques and systems we will access this market with a new product, which will bring benefits for the whole value chain (from manufacturing to operation). Verdict'slatest data shows that the totalvalue of fuelsoldat service stationsacross Europe (based on 29 European countries) will be close to **€600bn by the end of 2018**, as a result of high fuelprices and a marginal rise in fuelvolumes. From a studyconducted by STI/ SPFA – Steel Tank Institute/ Steel PlateFabricatorsAssociation[8] on **typical tank purchase by station owners** a typical store own two 12,000 gallon tanks and one 20,000 gallon tank (44,000 gallons total).Some petroleum marketers anticipate fluctuating fuel prices and install more capacity so they can buy and store additional fuel when itis offered at lower prices—for example, two 20,000 gallon tanks, one 15,000 gallon tank, and one 12,000 gallon tank (67,000 gallonstotal).Small mom-and-pop stores install less capacity, perhaps two 12,000 gallon tanks (24,000 gallonstotal).In extremely rural areas, a single 20,000 gallon tank with multi-compartments might be installed.

Unit: Number of petrol stations

COUNTRY	END 2014	COUNTRY	END 2014
Austria	2 622	Italy	21 800
Belgium	3 386	Latvia	609
Bulgaria	2 980 **	Lithuania	
Croatia		Luxembourg	238
Cyprus	280 *	Malta	
Czech Republic	3 792	Netherlands	3 825
Denmark	2 007	Poland	6 479
Estonia	440 *	Portugal	2 709
Finland	1 892	Romania	2 050
France	11 356	Slovakia	857
Germany	14 562	Slovenia	546
Greece	6 245	Spain	10 712
Hungary	1 914	Sweden	2 723 *
Ireland	1 798	United Kingdom	8 609
EU TOTAL	114 431		
Norway	1 602		
Switzerland	3 480		
Turkey	12 623 *		
TOTAL NO + CH + TR	17 705		
TOTAL	132 136		

EU * Numbers for 2013

Figure 12 Targeted Filling station by EU countries

With these data we can estimate:

a) a total of: 132136 (petrol stations in EU) x 3 tanks (2X 12,000 gallon tanks + 1X 20,000 gallon) = **390,000** tanks in EU

b) a total of: 10,234 (unmanned petrol station in EU) x 3 tanks = **30,702** tanks inside unmanned petrol stations (growing market)

Primary Targeted Market	Total Market Size	Growth Rate	Market Trends
EU Filling Stations	132316	3.5%	Increasing trend due to high demand of transportation vehicles and logistics

Main Selling Point:

Main selling points are based on following features for customers:

- Measures and displays on-line AE activity
- 4" touch sensing display (not needed here as data will be transmitted wirelessly)
- Integrated AE sensor signal conditioning circuitry
- 8GB storage space on internal SD card
- Wireless data downloading capability
- Long-term, unattended operation possible by providing power through a DC supply connected to mains or PV module
- Low cost (4655 euros price)

[7] Frost&Sullivan, 2015

[8] http://www.atlasoil.com/Blog/Fuel-storage-capacity-at-service-stations

Most Relevant Segment for initial introduction of new solution:

Segmentation, Targeting, Positioning (STP) model of MoniTank, Targeted market represent share we plan to capture in each of customer group:

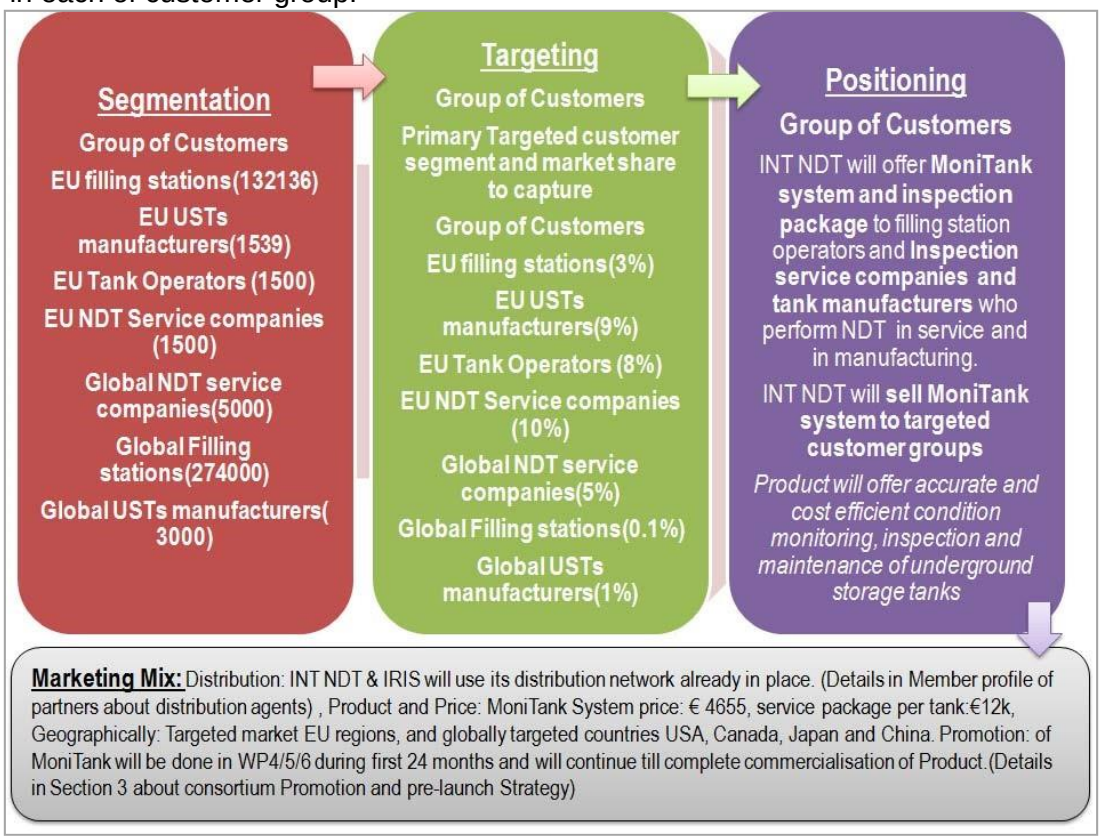
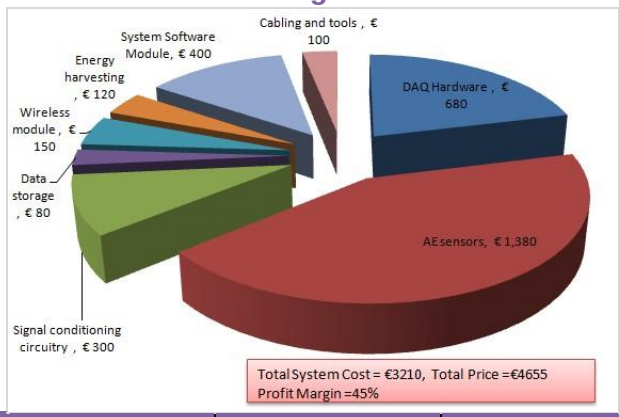


Figure 13 STP Model and Marketing Mix of MoniTank

Primary and Secondary Targeted Market and Market share in 5 years of post-commercialisation:

Table 15: Targeted Market



Primary Targeted Market	Size (No. Of companies)	Market Share (For MoniTank)	Number of Targeted Customers
EU Petrol/Fuel Stations	132136	3%	3967
EU Underground storage tanks manufacturers	1539	9%	133
EU Tanks operators/storage production facilities	1500	8%	115
EU NDT Service Companies	1500	10%	157
Secondary Targeted Market	**Size**	**Market Share**	**Targeted Customers**
Global NDT Service Companies	5000	5%	250

Global Petrol/Fuel Stations and t anks operators (USA, Canada, China, Japan)	274000	0.1%	274
Global underground storage tank s manufacturers	3000	1%	30

Scientific and Technological Impact of the Project:

The expected impacts in on technology, socially and environmental are:

- **Technical**: The commercial deployment of MoniTank System represents a major technical achievement that will lay the foundations for new technology developments e.g. monitoring and risk assessment of pressure vessels and structural integrity of underground storage assets.
- **Social**: The main social benefits will be in jobs creation (**112 new jobs** within the consortium) and in improved safety of operation of the underground storage operations.
- **Environmental**: The direct environmental impact will be to Figure 14 Cost Breakdown of MoniTank System improve the safety and reliability of fuel stations reducing the occurrence or likelihood of environmentally damaging leakages. MoniTank will contribute to the overall reliability and safety of petrol/fuel stations, thus enhancing the prospects for safe sustainable storage operations without disruption or loss of fuel due to structural failures.

Pricing Strategy: is depicted in figure hereby with breakdown of cost, price and profit margin.

Economic Relevance of the proposed Solution and Financial projections of consortium Members:

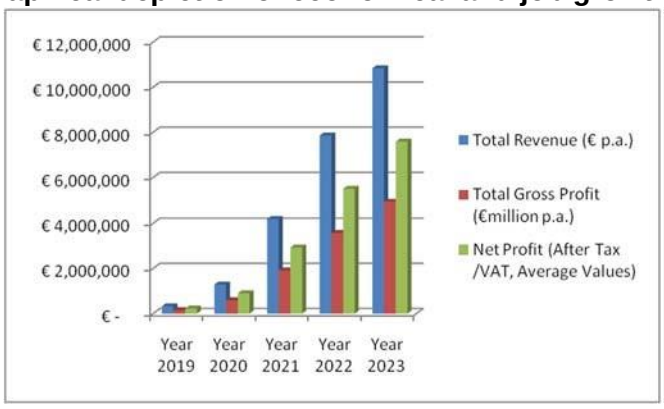
Table 16: Financial forecasting of product sales and partners turnover

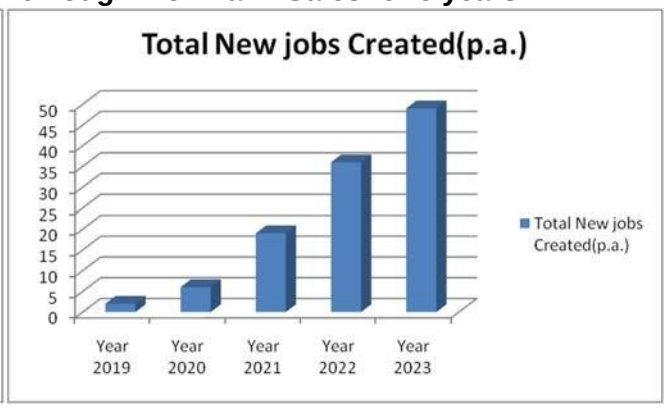
Forecast	Year 2019	Year 2020	Year 2021	Year 2022	Year 2023	total
MoniTank Product Sales Price (1% inflation rate is considered in yearly pricing)						
MoniTank System **(€)**	**4655**	**4701**	**4747**	**4793**	**4839**	
MoniTank Service Package **(€) per tank**	**12000**	**12120**	**12241**	**12364**	**12487**	
Expected Sales to Targeted Customers						
EU Petrol/Fuel Stations	26	132	661	1323	1825	3967
EU Underground storage tanks manufacturers	6	15	25	39	48	133
EU Tanks operators/storage production facilities	8	16	25	27	39	115
EU NDT Service Companies	7	20	30	48	52	157

Forecast	Year 2019	Year 2020	Year 2021	Year 2022	Year 2023	total
Global NDT Service Companies	8	28	38	78	98	250
Global Petrol/Fuel Stations and tanks operators	2	36	59	75	102	274
Global underground storage tanks manufacturers	0	2	5	8	15	30
Total Potential customers	**57**	**249**	**843**	**1598**	**2179**	**4926**
Total MoniTank Unit Sales (in €)	228,095	1,095,333	3,883,046	7,529,803	10,355,460	23,091,737
Total MoniTank Systems + Services Sales(in €)	96,000	193,920	306,030	333,818	487,003	1,416,770
Total Revenue (€ p.a.)	324,095	1,289,253	4,189,076	7,863,621	10,842,463	24,508,507
Profit Margin= 45% on System and 55% on system+ service package, Cost of Unit = €3210						

Total Gross Profit (€million p.a.)	155442	599555	1915687	3572010	4927808	**11170505**
Total Annual Cumulative ProfitInitial Investment	-2144557	-1545001	370685	3942696	8870505	
Net Profit (After Tax /VAT, Average Values)	108809	419689	1340981	2500407	3449465	**7819353**
Job creation estimation (1 job per 100k profit)						
Total New jobs Created(p.a.)	2	6	19	36	49	**112**
Turnover of consortium Partners (Values in € p.a.)						
Integrity NDT(70% profit share)	108809	419689	1340981	2500407	3449465	**7819353**
RIS(20% profit share)Software License	15544	59955	191568	357201	492780	**1117050**
BUL(5% profit share)	12435	47964	153254	285760	394224	**893640**
TWI(5% profit share)	12435	47964	153254	285760	394224	**893640**
FLOTEKS as end user and consumer will be given 5**%** discount on MoniTank System, and will have access to non-commercial exploitation of MoniTank demos and development results						
New Jobs created per partner	INT NDT(**78**), IRIS(**11**), BUL(**9**), TWI(**9**), Floteks (**5**)					
Return on Investment for EC considering 2.3 million budget						
Return of Investment for EC	**11:1**	in 5 years Post commercialisation				
Return on Investment for consortium on 5 years of sales of MoniTank						
Return of investment for on sales of MoniTank Units	**15:1**	For Consortium, calculated considering consortium investment and net profit in 5 years,				

Graphical depiction of economical and job growth through MoniTank Sales for 5 years





<u>Capital Investment Plan for three years of operation and Funding Mix and relevant Project Activities</u>:

- Total project eligible costs including FTI funding (€2 million) + consortium partner internal cost input (€500k) = €2.5million
- Further funding requirement for first 2 year period following project completion from internal partner funding/bank facility/venture capital and income inflow from licensing + support fees = €0.5million.
- Sales of product and associated services to tanks operators will yield above average income returns due to product/service uniqueness and attractiveness to the end-users based on benefits on all fronts, i.e. economic, safety, environmental and societal.

<u>Barriers/obstacles/framework Condition and MoniTank Strategy to address:</u>

Table 17: Barriers and mitigation plan

Barriers/obstacles and framework conditions	MoniTank strategy to address barriers
State of the art methodologies of visual inspection and other NDT techniques	MoniTank through effective communication, marketing and clients engagement will demonstrate the efficiency of MoniTank and will gain confidence of end users in investing in innovative monitoring system for millions of savings(as calculated above)
Certification and Standardisation requirement for such monitoring and inspection products	Certification and standardisation process will start in early stage to have approved quality software in compliance with plant operation and maintenance. Storage tanks/pressure vessel legislative requirements and protocols will be addressed in MoniTank as well.

Table 18: PESTLE Analysis: shows dynamics of market of MoniTank

POLITICAL

- The involvement of partners from different EU countries will result in a widespread application of the project's technology which means all countries will benefit.
- Fossil fuel consumption forecast to increase until 2050, increasing demand for safe storage. Liquid and gas fuels (biofuels, hydrogen) will remain important particularly in the transport sector.

Integrity NDT will provide a guideline to oil companies and local authorities in determining the safe boundary of petrol station

ECONOMICAL

- Enhanced competitiveness in the European and global market (USA, ASIA, etc.)
- Enhanced competitiveness in the EU growing market of unmanned/virtual assisted petrol stations

SOCIO-CULTURAL

- Reduction of risks and accidents related to leakages and vapour emissions, improved safety of petrol stations (workers, citizens)
- Improve confidence in petrol stations security and in particular in unmanned/virtual assisted stations
- Reduced need for internal inspections improves worker safety and reduces pollution.
- New High skill jobs in the EU.

TECHNOLOGICAL

- High performance, innovative and low cost monitoring system for USTs, providing real time data on the structural integrity of the tanks
- New methodology based on Risk Based Assessment(RBA) + Structural Health Monitoring(SHM) + Quantitative Risk Assessment(QRA)
- Predictive maintenance of USTs

LEGAL/LEGISLATION

The leak detection standards (EN 13160 1-7): Leak detection systems. General requirements and test methods for interstitial spaces, leak protecting linings and leak protecting jackets

The Water Quality Framework Directive (WQFD) adopted by European Parliament on July 18th 2000.

AE standards:

- ASTM E 1930 Standard Practice for Examination of Liquid-Filled Atmospheric and Low-Pressure Metal Storage Tanks Using Acoustic Emission.
- ASTM E 1211 Standard Practice for Leak Detection and Location Using Surface-Mounted Acoustic Emission Sensors.
- BS EN 15856 Non-Destructive Testing. Acoustic Emission. General Principles of AE Testing For the Detection of Corrosion within Metallic Surrounding Filled With Liquid.
- HPIJ, A Recommended Practice for Acoustic Emission Testing For Corrosion in the Bottom Plate of Aboveground Tanks, 2005.
- BS EN 14584 Non-Destructive Testing. Acoustic Emission. Examination of Metallic Pressure Equipment during proof.

ENVIRONMENTAL

- Minimising the environmental impact and sustainability of human activity.
- Reduction of risks of soil and water supplies pollution, environmental damages caused by fire and explosions.

2.2 Measures to maximise impact

a) Dissemination and exploitation of results

Dissemination and Exploitation Plan:

MoniTank consortium is already working with European NDT companies (the problem solvers), and the storage tanks/pressure vessels manufacturers and tanks operators (the problem owners). They have provided us with considerable information about their requirements and expectations of our system. We will continue this dialogue and introduce, MoniTank as a potential solution for monitoring of underground pressure vessels. Our demonstrations at end-user sites FLOTEKS will be critical to provide confidence that investment in the MoniTank will provide a return on their investments. The decision-making processes require high levels of proof; one of the outcomes of MoniTank will be the development of the investment cases for users, tailored to their approach. The market analysis for MoniTank has been and will continuously be performed initially in EU countries. MoniTank has the commercial potential to reach global markets, mainly the North America and Asia Pacific; these markets will be approached in first phase and rest of the world at a later stage, focusing on critical success factors for deployment.

Credible path of Innovation delivery to Market:

Supply chain and Business Model of MoniTank is shown in figure 15, 16. **Dissemination and exploitation measures:**

INT NDT as coordinator will lead the development of the exploitation and commercialisation strategy with commercial partner IRIS and other members; it will be based on the following activities (Activities to be implemented in WP5):

Dissemination Measures:

The project will take a pragmatic approach to the **dissemination of**

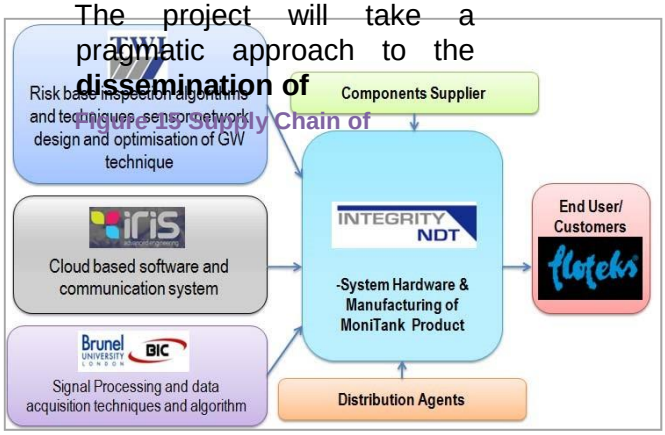
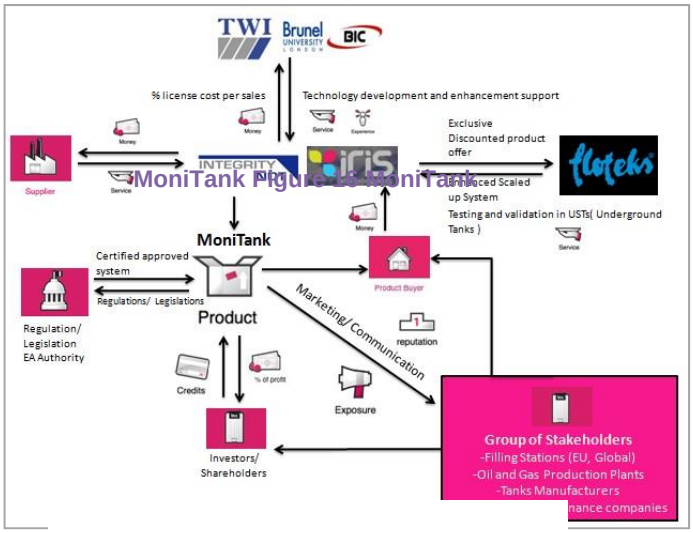


Figure 15 Supply Chain of MoniTank



MoniTank Figure 16 MoniTank Business Model

MoniTank, as it is seen as central to success. The main dissemination activities will be undertaken by INT NDT (in **WP5**) as mentioned below:

- MoniTank project **website** will be the most important dissemination channel during the project. The MoniTank Web presence will serve as a central point of entry for all public materials. This site will also be integrated with social media. Marketing tools will be used to direct potential clients to the website.

- **Participation in EU and global activities** such as NDT technical groups, conferences(e.g. World Conference on NonDestructive Testing(WCNDT),BI NDT,ECNDT 2017,ASNT 2018 etc.) and concentration meetings(pressure vessels Group)
- **Cooperation with other projects and initiatives** will bring valuable expertise and resources to the project, and will also be a key opportunity for advertising the MoniTank activities.
- **Technical papers and Trade Journals** will be published in conference proceedings and journals. Special emphasis will be given to publishing in peer-reviewed reputable journals and trade journals.
- By means of **newsletters and project innovation briefs**, the project will reach out to a wide audience and contribute to relevant EU policy developments in the area of international cooperation for research and innovation.
- **Publication of project results** in trade journals and international conferences (such as European NDT Conference in Europe and the International Conference of NDT).
- Preparing **marketing material** including project brochures directed both towards potential users and to other relevant organisations, e.g. Global and the European Institutes of NDT and European standards organisations.

Exploitation Measures:

- **Route-to-Market:** The customers will initially be storage tanks manufacturers, fuel station operators and inspection companies) and NDT services systems suppliers. As a product and inspection services supplier company(INT NDT), we will directly approach customers with their stations operating USTs in EU mainly; *Shell(14,000), Total(11,500), BP(8,900), Esso (7,100), ENI(6,400), Orlen (2,800), AVIA(2,600), OMV(2,500), Lukoil (2,500),Statoil(2,000).*
- **Exploitation and Business models:** There are several ways to ensure a positive return on investment from the MoniTank and its IP. However, we want to ensure maximum dissemination and exploitation of the technology without compromising our overall IP position or revenue potential. The development of a flexible licence model is vital for this. This **approach** would encompass: *(i) An IP licence for market segment and territory, (ii) Regional supplier licenses for franchise operators of the MoniTank solution, (iii) licensed benchmarking and optimisation consultancies.*
- **Marketing and sales:** We will identify our first customers, many of whom we are already in discussion with in Tier 1 market. We expect to license the sales of the software to agents in North America, Asia Pacific and US beside direct sales in EU.
- **Agents and Distributors:** INT NDT, IRIS and TWI distributors and agents will be engaged to enable sales of MoniTank, agents list is detailed in section 4.

Table 19: Project potential uses and relevance

Potential Users and uses	MoniTank Relevance
Technological Sector and Research Community	MoniTank provides research community knowledge being generated through the development of RBI algorithms and bringing applied research to commercial usage via MoniTank offering NDT research community progress beyond state of the art monitoring tool for tanks.
Commercial	Commercialization of MoniTank will result in potential benefit for users including maintenance service provider companies, fuel stations operators, tank manufacturers and process industry as it will directly impact by reducing maintenance cost, downtime, decommissioning due to maintenance, structural failures and leakages offering overall cost efficient and safe environment by doing preventive real time monitoring of underground storage tanks. MoniTank consortium will obtain **€24 million in sales of MoniTank in first five years of commercialisation.**
Social	MoniTank system offers a profit margin of **45%** thus providing an attractive opportunity for investors to take economic benefit by investing in such an innovative technology.
Environmental	Contribution to safety by reducing possibility of accidents due to structural failure, fatigue in pipes thus creating safer workplace and environment for workers.

Policy Making and Standards	MoniTank will provide inputs to industrial standards legislations, emerging standards, policies regarding inspection protocols and needs of regular monitoring of heavy equipment piping system.
Skills and Educational Training	MoniTank provides research community knowledge being generated through the development of RBI algorithms and bringing applied research to commercial usage via MoniTank offering NDT research community progress beyond state of the art monitoring tool.

Key Stakeholders for commercial exploitation: External stakeholders to be involved in the latter end commercial exploitation of the project innovation will include EU and overseas Trade and Industry Associations, EU and global Chambers of Commerce and overseas trade delegations, fuel stations associations from EU countries. NDT inspection companies will be potential stakeholders for commercial exploitation of MoniTank System as well. Floteks will play a key role in engaging their group companies globally for successful market penetration of MoniTank. Integrity NDT will engage its distribution network for successful exploitation.

TWI has clients in drilling, production, storage and overland piping facility operators and owners. (as mentioned in members list, www.twi-global.com/membership/our-industrial-members),**BUL** is providing research and technology development services to oil industry, **Integrity NDT** is NDT service and product development SME active on many sectors including tanks inspection and monitoring, **IRIS** is facilitating digital system for monitoring application in production industry. **Floteks** is tank manufacturer having distribution and sales of tanks all across EU. All consortium members of MoniTank are providing services or products to the supply chain companies of inspection of tanks, targeted in MoniTank, thus will leverage the network for engaging clients and sales of MoniTank.

Table 20: Competitiveness and Innovation capacity development of Consortium Members

Partners	Corporate Strategy	Benefit to the competitiveness and Innovation Capacity
INT NDT	A specialist provider of innovative inspection services and solutions, our quality is highly valued by ourselves and our client customers. From small local manufacturers to international companies worldwide, we are committed to delivering cost effective Inspection Solutions and Technical Assurance of the integrity of plant equipment and fabrications to our clients, regulatory and insurance bodies alike. Our quality systems are underpinned by ISO9001 certification.	MoniTank system development and commercialisation will enable INT NDT company to expand its product and services portfolio, enhance its revenues, create internally more jobs and expand new markets and regions for sales of MoniTank system.
IRIS	To improve industry by creating and integrating profitable high-end technologies and solutions.	MoniTank will directly impact competitiveness of IRIS as software and communication module being optimised by IRIS will create return on investment as well as will enhance product competencies and knowledge of condition monitoring applications.
TWI	TWI is one of the world's foremost independent research and technology organisations, with expertise in materials	MoniTank system will expand the inspection systems and services portfolio of TWI; will strengthen their
Partners	**Corporate Strategy**	**Benefit to the competitiveness and Innovation Capacity**
	joining and engineering processes as applied in industry. TWI aims to be the leader in innovation, knowledge transfer and in solving problems across all aspects of manufacturing, fabrication and whole-life integrity management.	expertise in investigating and monitoring critical assets of ageing storage tanks. It will benefit TWI to enhance their services of inspection as well as fuel stations integrity.

BUL	Brunel Innovation centre has been dedicated to close the gap between academic research and industrialisation of technology. BUL is dedicated to technology transfer and industrial impact of research.	BUL utilises its expertise in NDT systems, development of algorithm and software platforms and will expand its resources, experiences and staff to have more provision of development contracts as innovation centre and technology developer.
Floteks	To provide high quality product, to manufacture products in house and export worldwide, expand markets and business through in house product development and continuous innovation of high quality standards	MoniTank system application on underground storage tanks being manufactured by FLOTEKS will help them to enhance its life by integrating such monitoring system with their tanks; we will be able to offer tanks with condition monitoring system another revenue stream.

Beside activities already explained other steps before taking MoniTank to market will be:

- **IP protection of results/foregrou** Technological development and sc up systems, algorithms, sensors, system architecture will be patente before commercialising the system
- **Production Plan** In order to reach targeted cost, manufacturing proce will be optimised and cost efficient components selection will be made Further, production partner agreem will be secured.
- **Business Development Team Training** Business development, sa team and distribution representativ INT NDT will be trained about proc features and applications to facilita marketing and client's awareness.
- **Product Initial Introduction in Ev and Marketing** INT NDT will attend NDT exhibitions and relevant stakeholders' events for initial feedback, response and interest of potential clients in MoniTank system. B2B meetings will be organised with interesting stakeholders to obtain customer feedback and purchase interest as well.

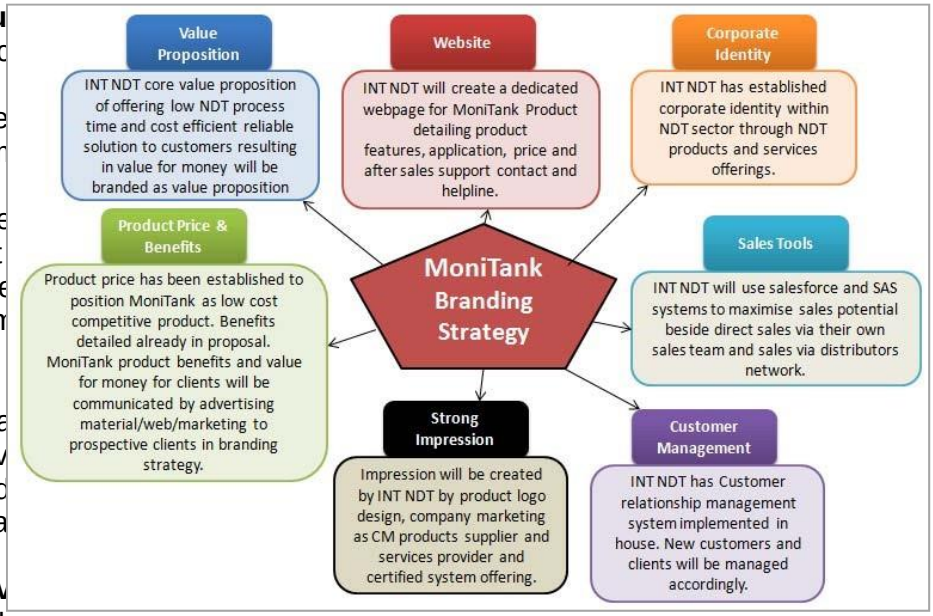


Figure 17 MoniTank Branding Strategy

- **Distributor's agreements finalisation:** INT NDT will formulate agreements with existing and new distributors globally for MoniTank.
- **Product Packaging:** INT NDT will **develop and fabricate intrinsically safe enclosure/box** required for electronic devices for packaging of MoniTank System.
- **Product Branding:** INT NDT will develop and implement **MoniTank product branding strategy** as shown in figure below.

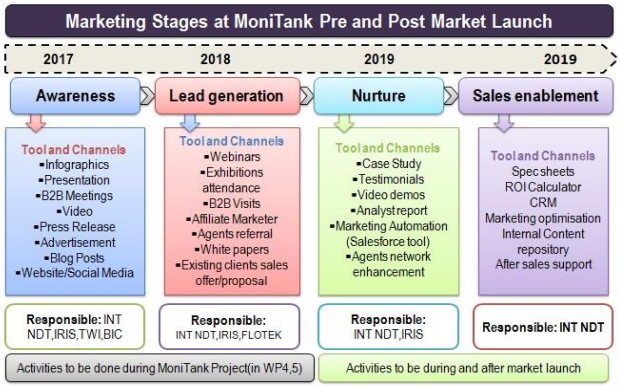


Figure 18 MoniTank Marketing and Sales Effort Plan

Market Positioning, commercial strategy and Marketing/Sales Efforts: planned strategy is depicted in fig 18.

Table 21: Future Competitiveness Strategy

Expected Results	Further Development Strategy	Measures of Future Competitiveness
MoniTank structural health monitoring system for underground storage tanks	The system will be further, developed to extend its application to above ground storage tanks, connected pipes, vessels and valves and all other equipment under different environmental conditions. The second generation system will be IoT enabled and cloud based and will be launched after 5 years of commercialising and sales of first version of MoniTank product.	MoniTank product will be optimised regularly as per customer feedback and continuous upgradation and performance improvements will be done in the product for sustainable competitive advantage.

Other Measures to enhance Impact: In addition to the full suite of dissemination activities as previously described, the consortium will be publicising information surround the MoniTank system, its developments and the project results through a number of industry specific journals and publications. **Examples of these publications include**, but would not be limited to: NDT Magazines, Industrial Engineering, Future tanks storage Technologies, Computerworld, Global Trading Magazine, IEEE journals. During the course of this action, all dissemination activities undertaken will focus on the advancement of the MoniTank system from TRL6 to TRL9. Once the action is complete and the MoniTank system is commercialised, the ongoing dissemination activities will be targeted at enhancing exposure of the product to a wider market. These actions will include but not be limited to ***(i)*** Agent demonstrations in Non-EU territories ***(ii)*** Direct engagement with petrol station Operators ***(iii)*** Case studies of MoniTank system and its operational results **(iv)** Downloadable content through the MoniTank website.

b) Intellectual Property, knowledge protection and regulatory issues

IPR Management: Specified IP required for sell the MoniTank product will be with Integrity NDT. This provides freedom for Integrity NDT to operate in the underground storage tanks market and to access capital required for further growth. An initial patent search has been conducted which indicates that there are no infringement issues with other IP. This will be updated during the course of the action. IP generated during the action will reside with the company generating the IP, and where necessary for commercial exploitation of MoniTank.

These IP arrangements are summarised in Table below:

Table 22: IPR Management

Background IP	Ownership	Freedom to Operate	Foreground IP	Ownership
Systems Integration, instrumentation	Integrity NDT	Freedom to operate for project	Manufacturing process of complete MoniTank system (patent)	Integrity

		development work		NDT Will own the complete system and will share profit as per agreed in project consortium and IP ownership agreement,
AE sensors configuration, know-how on CM and sensor deployment, signal and image processing of AE-GW techniques, pattern recognition algorithms, electronic design of systems for AE and vibration analysis.	TWI	Freedom to operate for project development work	Integrated system of MoniTank (patent)	
Signal processing and data analysis	BUL	Freedom to operate for project development work	Sensors validation, machine learning tool, training material and standards	
Software, communication and cloud based system	IRIS	Freedom to operate for project development work	Copyright of User interface software and data analysis algorithms	

Measures to ensure Commercial Exploitation:

In order to ensure commercial exploitation of developed IPs, the most important task will be to protect the developed system via patent before exploitation; this will be done by coordinator with the support of other partners. Other measures will include B2B meetings with potential clients, marketing of product in regions of interest and establishing a strong network of distribution agents and affiliate marketer to engaged potential clients and boost product sales. Fig 19 shows SWOT of MoniTank where strengths and opportunities will be exploited to cater external threats and weaknesses will be countered by mitigation plan as proposed in section 3.

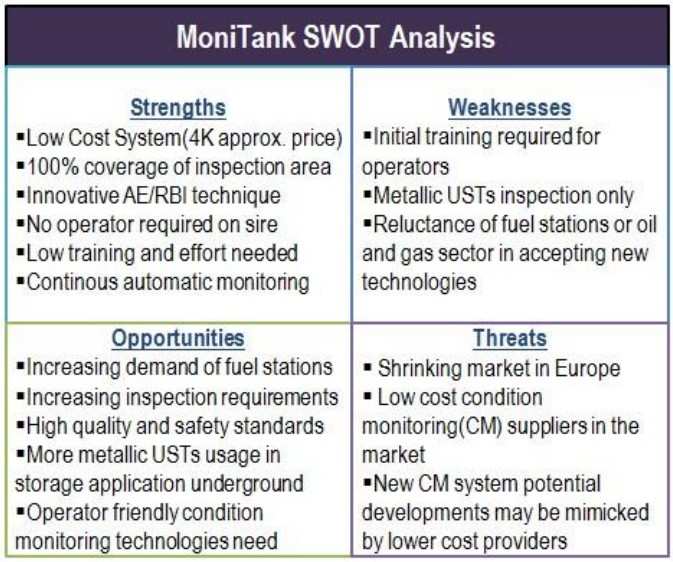


Figure 19 MoniTank SWOT Analysis

Knowledge Management and Protection Strategy:

Publishing scientific papers about the results is also one way to be used for MoniTank dissemination. All the publications will be identified as the result of research of the project, which is supported by the European Commission (Horizon 2020 – SME instrument). General dissemination activities will include (i)

the inclusion of project results on the partner's web sites. (ii) The publication of project results in technical papers, trade journals and conferences. (iii) Disclosure of information through project brochure to relevant associations (e.g. The National, European NDT associations and/ or SHM Users Associations). The partners will publish papers on the application of MoniTank technology.

Scientific knowledge and open access:

With regards to scientific publications, the Project coordinator will compile all the papers draft and the indication from the authors about the selected journals for publication. Coordinator will identify and propose green access solutions for all the scientific papers that will be written. Whether it will not be possible to reach the gold open access standard, the PC will define with the authors of the paper a good scientific journal that will allow the highest visibility possible. Besides, the PC in associations with the External stakeholders Board, will be engaged to use the green open access standard (e.g. USIEGEN), in order to have an online repository of the manuscripts. If the already existing academic repository for green open access will not be sufficient, the PC will decide to create a specific USIEGEN repository following the green open access standards.

Regulations/Standards Requirements before commercial exploitation:

Table 23: Relevant Directives to MoniTank

EC Directives	Details
Council Directive 96/82/EC1 (will become 2012/18 from 1st June 2015)	MoniTank will contribute to managing the risk of operating process plants as covered in this Directive on the Control of Major Accident Hazards involving dangerous substances (The COMAH Directive).
Directive 96/62/EC, part of 2008/50/EC	MoniTank will support this EU legislation on pollution control as it proposes to eliminate the requirement to empty and clean the storage tanks that releases huge amounts of hazardous fumes into the atmosphere.
Directive 1999/92/EC, with later amendments	MoniTank will support this Directive setting the European minimum requirements for improving the safety and health protection of workers potentially at risk from explosive atmospheres.
Directive 89/391/EEC, with later amendments	Also known as the Framework Directive that introduced measures to encourage improvements in the safety & health of workers at work will also be supported by MoniTank.
Directive 2000/60/EC, with later amendments	This Directive established a framework for community action in the field of leakage of oil and the project will achieve the latter by offering more reliable storage tank inspection enabling the avoidance of hazardous liquid leakage into the ground.

c) Communication

With regard to external communication, in order to ensure a maximum outreach and impact of the MoniTank project, we will follow a multi-channel dissemination strategy in order to gain maximum visibility including academia, the industry, welfare and end-user organizations, policy makers, and the general public. These activities will be covered in **WP5**.

The communication strategy will focus on:

- Distributing relevant results among stakeholders and potential customers
- Propagating research results in the form of new technologies, prototypes, and interactive concepts to the respective industry branches and public and corporate audiences;
- Organizing seminars and workshops to disseminate knowledge on the methodology and technology used and the results obtained in the project to professional and non-professional social inclusion actors;
- Informing the general public of the project results through a dynamic website with high-quality content explaining the idea of the system and the guidelines for the evaluation of technology; and
- Development of an appropriate social media strategy aimed at academics; potential users of MoniTank and related third parties such as journalists, bloggers etc.

TWI and BUL will publish and present the main publishable technical findings in refereed high impact journals, proceedings of international conferences and workshops, such as:

- EUROCORR (Annual Conference organised by the EFC), NDT&E International

- World and European Conference on Non Destructive Testing
- International Journal of Condition Monitoring (BINDT publication)
- Conferences of American Society of Non Destructive Testing

3. Implementation

3.1 Work plan — Work packages, deliverables and milestones

Brief presentation of the overall structure of the work plan: MoniTank work plan has been developed to maximise the effectiveness of commercialisation. PERT diagram represents different work packages and their inter link. Previous development of MoniTank system will feed input to this project to further, scale up the technology already at TRL6 to TRL 9. **WP1** will deal with scaling up hardware, software, and AE sensor network and communication system as per commercial requirement. This scale up system will be tested in underground storage tanks in **WP2** and continuously performance will be evaluated and data will be analysed in order to reach the optimum required commercial performance. The final validated system will be commercial ready in **WP3** with required packaging and certification. For production and creating volumes of MoniTank for sales, manufacturing and components procurement plan will be established in **WP4** to minimise the production cost and to meet the future demand of system.**WP5** will deal with dissemination, exploitation and communication activities of the project facilitating the commercialisation of project and maximising the commercial targeted impact of developed product. In **WP6** marketing, sales and distribution network activities will be planned for pre-launch and post launch of product planning. **WP7** will finally deal with product management, coordination, risk management and innovation management activities of the project.

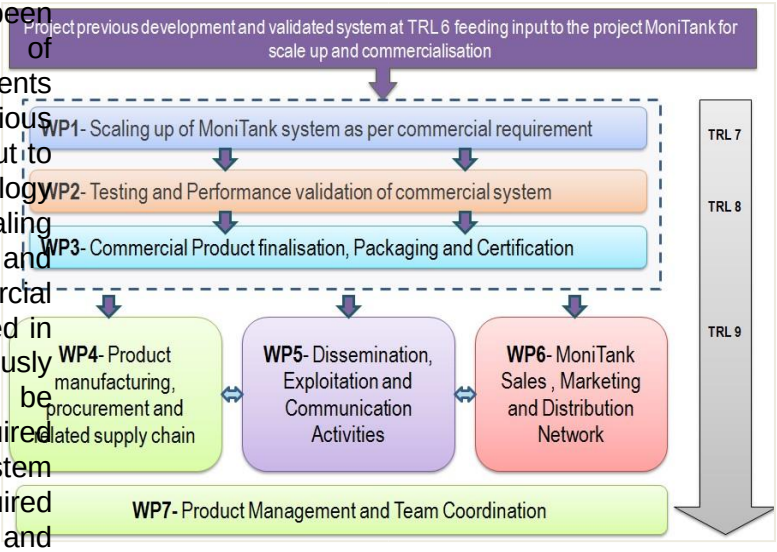


Figure 20 MoniTank Project PERT

Gantt chart of the Project:

MoniTank	Leaders	Project Months																													
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
WP1 - Scaling up of Monitank system as per commercial requirement	**INT NDT**																														
T1.1 Scaling up MoniTank system hardware to commercial version	INT NDT																		**D1.1 & M1**												
T1.2 Optimisation of Software and GUI Interface	BUL																		D1.2												
T1.3 Scaled up system trials	INT NDT																		D1.3												
WP2 - Testing and Performance validation of commercial system	**Floteks**																														
T2.1 System testing in factory and industrial environment	Floteks																				D2.1										
T2.2 Performance verification and system validation	TWI																				D2.2										
T2.3 Optimisation and customization of complete commercial product	INT NDT																						**M2**								
WP3 - Commercial Product finalisation, Packaging and Certification	**INT NDT**																														
T3.1 Finalisation of MoniTank's components and packaging	IRIS																														
T3.2 Final Commercial market ready MoniTank Product	INT NDT																												**D3.1 & M3**		
T3.3 MoniTank certification, standards compliance and training development	TWI																													D3.2	
WP4 - Product manufacturing, procurement and related supply chain	**INT NDT**																														
T4.1 MoniTank Product manufacturing and components procurement plan	IRIS																														**D.4.1 & M4**
T4.2 System supply chain finalisation and related agreements	INT NDT																														
WP5 - Dissemination, Exploitation and Communication Activities	**INT NDT**																														
T5.1 Dissemination and exploitation plan and IP management	INT NDT			D5.1																								D5.2			**M5**
T5.2 Communication and Advertising including demos development	IRIS																														
T5.3 Awareness Events and Promotion	BUL																														
T5.4 MoniTank pre-launch activities and clients engagement	INT NDT																														D5.3
WP6 - Monitank Sales, Marketing and Distribution Network	**INT NDT**																														
T6.1 Monitank sales and marketing plan	INT NDT																													**D6.1 & M6**	
T6.2 Distribution sales agents network establishment	INT NDT																													D6.2	
T6.3 After sales support services and maintenance procedures development	IRIS																														
T6.4 Business development and product support team training	INT NDT																														
WP7 - Product management and Team Coordination	**INT NDT**																														
T7.1 Financial and administrative management	INT NDT																														
T7.2 Risk and innovation management	BUL																														
T7.3 Product development Reporting and progress meetings	INT NDT																														
T7.4 Quality assurance and Performance monitoring	INT NDT																														
First year report	**INT NDT**																		**D7.1**												
Project final report	**INT NDT**																														**D7.2**
Risk management report	**BUL**																														**D7.3**

3.2 Management structure and procedures

Management Structure of Project:

Organisation structure and decision making

Overall structure: The management structure proposed for the MoniTank aims at facilitating the cooperation between partners while maintaining a strict control of the gradual achievements of the project objectives. It distinguishes between decision-making structures and organization of daily operations. This aspect, as well as the relevant bodies, and specially their composition, are described below in further detail. In order to enable a management structure that is both effective and lean, the structure provided in Figure will be implemented.

Project Coordinator: The Coordinator, **Mr. John Stephenson, Integrity NDT** will monitor continuously the scientific project progress. He will review of scientific reports and deliverables before they are uploaded in the project's intranet and EC portal. Any progress or alteration of the foreseen schedule will be reported through a continuous and fluid communication, and to the partners, through half-yearly meetings to evaluate the Project's development and the eventual correcting measures. He will manage overall administrative and financial tasks of the project.

Technical Manager: The Technical Manager, **Ms. Ángela Angulo, TWI** will support the coordinator with the continuous monitoring of the technical project progress. He will review project reports and deliverables before sending them to the coordinator. Any progress or alteration of the foreseen schedule will be reported to the Coordinator, through a continuous and fluid communication. The Technical manager will keep a fluid communication with the work packages leaders by phone, e-mail, video-conference or physical meetings.

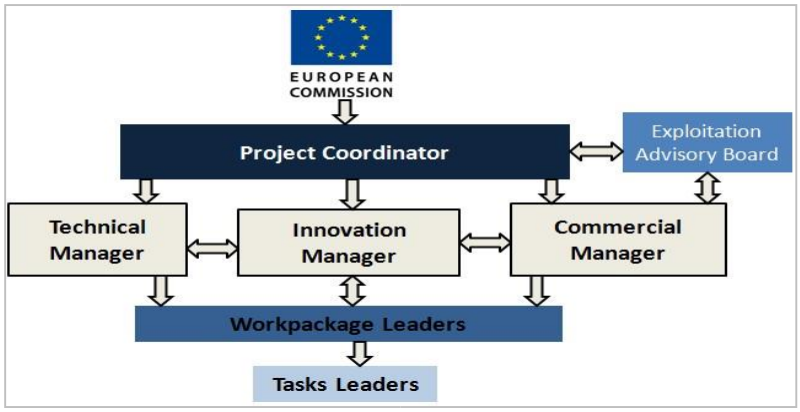


Figure 21 Management Model of Project

Commercialisation Manager : **Mr. John Stephenson, Integrity NDT** will manage the exploitation and commercialisation of the project. He will be in charge of developing and implementing an effective commercialisation route, customers' network and knowledge management strategy for the effective market take up of MoniTank product. Product commercialisation as a core focus of the project will be addressed as an agenda item during the meeting of the project consortium.

Innovation Manager: **Dr. Edurne Gastón, IRIS** has been selected carefully as responsible for overlooking both technological and commercial developments in MoniTank. She will facilitate the project coordinator, technical and commercial manager by providing insights about market and technical problems, with a goal of successfully implementing appropriate creative ideas. She will overlook market opportunities and product needs to ensure high performance improved MoniTank product.

Progress monitoring: Progress meetings will be held on a regular basis. Each partner will report to the WP leaders and to the Coordinator about the progress of activities. The reports will include information about the innovation progress and results obtained and will be reviewed before submission to EC. Deliverables and milestones will be reviewed at each meeting. The project website private area will be used as a tool to monitor the progress and also to provide the partners with the necessary information on the project (reports, results, conferences).

Regular Meetings Calendar: will be organised every 6 month to meet and check the project progress.

Communication strategy: Proper internal communication is essential. Informal day-to-day communication will be done by e-mail and telephone mainly. Teleconferences will be organized to discuss particular issues. Additionally, the internal part of the project website will be used, among other things, to track project progress, to make available continuously updated information about the progress of each WP, to exchange documents, to collect financial data, to collaboratively prepare reports. It will include project handbook describing the project plan, contact lists, reporting templates and Project guidelines. BUL will be in charge of updating website public contents according to the project progress. All information circulated will be treated as consortium confidential unless stated otherwise.

Quality Assurance: The quality control will be important to check that all contributions and conclusions are consistent and meet the requirements for the deliverables, reports and other documents.

3.3Consortium as a whole

Consortium members and Complimentarity of activities in value chain of MoniTank is depicted in figure below:

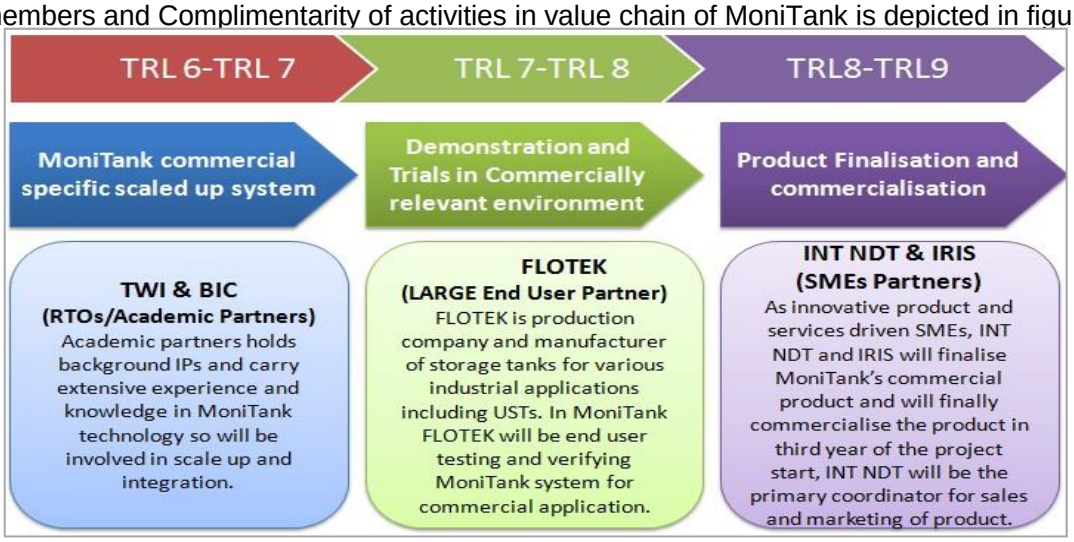


Figure 22 Partners Complimentarity

3.4